

Build a Five Wire Harness SOP (1x Episode: 5 Wires, 3 Ties)

One episode builds **one** harness from one five-wire set. The table begins with the **layout board** on the assembly spot, just right of the middle of the table. Three **tie stations** are marked across the board — **S1** nearest the connector, **S2** at the middle of the run, **S3** just short of the **breakout mark**. One **wire set** lies on the **set spot**, on the left of the table or at the front-centre, as set out below. A **parts tray** at the front-right holds the cable ties, the mating half, and the tape roll. A **cutter stand** at the back-right holds the flush cutter. A **peg rail** stands along the right edge, at the back.

The order never changes: lay the set on the board, comb the five wires into one bundle, then tie S1, S2, and S3 in that order, then mate the connector, then tape the breakout, then hang the harness on the peg. Each station is finished — tie tensioned, tail cut flush — before the next station is started. No connector is mated until all three ties are on and checked. The tape does not go on until the connector is mated. The harness does not go on the peg until the wrap has been checked.

The right gripper does the primary work: it combs the wires, takes and threads every tie, tensions every tie, works the cutter, takes and pushes the mating half, works the tape roll, and takes the right half of the two-arm carry to the peg. The left gripper supports: it lays the set on the board, holds the bundle closed at each station, holds the connector body during the mate, holds the branch against the main run during the wrap, and takes the left half of the carry. The left gripper stays on the left of the board and never reaches across it.

The table is set up in one of three ways. Only the wire set moves; the layout board, the parts tray, the cutter stand, and the peg rail are in the same place in all three.

- **Config L1:** the wire set is at the back-left, to the left of the board and toward the back edge.
- **Config L2:** the wire set is at the front-left, to the left of the board and toward the front edge.
- **Config M:** the wire set is at the front-centre, in front of the board and clear of the parts tray.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the wire set starts at the back-left (**Config L1**), the front-left (**Config L2**) or the front-centre (**Config M**), always lying along the table with the connector to the left. One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the **left gripper** picks the wire set in all three configs — the front-centre is on its side by convention. No arm reaches across the table. There is no right-side config: the

right quarter holds the parts tray, the cutter stand, and the peg rail, all worked by the right gripper, so the set never starts there.

- **Fixed roles:** everything else is the same in all three — the board on the assembly spot, the right gripper combing, tying, cutting, mating, and taping, the left gripper holding, and the order of the steps.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the wire set on the set spot for this episode's config, the layout board just right of the middle with all three tie stations and the breakout mark, the parts tray at the front-right, the cutter stand at the back-right, and the peg rail along the right edge.
3. The top face of the layout board is in frame from above, so all five wires, the three station marks, the breakout mark, every tie head, and every cut tail can be seen.
4. The connector end of the board is in frame from the front, so the latch can be seen clicking over and the gap between the two housings can be read.
5. Both arms are at home with grippers open.
6. The table is bare apart from the layout board, the wire set, the parts tray, the cutter stand, the peg rail, and robot hardware.
7. The right arm reaches S1, S2, S3, the breakout mark, the parts tray, the cutter stand, and the pegs without stretching. The left arm reaches the set spot (all three configs) and the whole left side of the layout board without stretching.
8. The travel from the parts tray to the board, and from the board to the peg rail, is bare and level.

Materials checklist

1. One **layout board** lies flat on the **assembly spot**, just right of the middle of the table, **square** to the front edge, on a mat that keeps it from sliding.
2. The board carries four printed marks along the run, readable from above: **S1** nearest the connector end, **S2** at the middle, **S3** just short of the **breakout mark**, and the breakout mark itself.

3. One **wire set** lies on the **set spot** for this episode's config, connector to the left: five pre-terminated insulated wires, all running to one **end connector**, with **three wires carrying on as the main run** and **two branching off** at the breakout.
 - **Config L1**: back-left, to the left of the board
 - **Config L2**: front-left
 - **Config M**: front-centre, in front of the board, clear of the parts tray
4. Every wire is whole: no split insulation, no flattened kink, no cut strand. Every **free end** is clean.
5. The **end connector** is already terminated on the five wires, with all contacts straight, none pushed back, and its **latch** free to spring.
6. The **parts tray** at the front-right holds, in separate groups: **three cable ties** plus **two spares**, **one mating half**, and **one tape roll**. Each group has its own place in the tray.
7. Each **cable tie** is a one-piece strap, long enough to pass under the bundle and thread its own head, and it locks with a click when the tail is drawn through.
8. The **mating half** matches the end connector, with its keying feature and its latch window clean and unbent.
9. The **tape roll** is harness tape, tacky, and it tears or unrolls without stretching thin.
10. One **flush cutter** stands in the **cutter stand** at the back-right, handles up, jaws closed, and its jaws sit flat against a tie head.
11. The **peg rail** stands along the right edge at the back, with every peg clear and nothing hanging on it.
12. Keep the left side of the table clear apart from the wire set. The left gripper comes in from there onto the set spot and the board.
13. Before collection, confirm by hand that:
 - the five wires comb straight and lie as one round bundle with no wire fighting the others;
 - a tie passes under the bundle at each station and threads its own head;
 - the cutter takes a tail off level with the head face and leaves no stub;
 - the two connector halves push together square and the latch clicks;
 - the harness hangs on a peg by its S2 tie without sliding off.

Workspace layout

- **Set spot**: where the wire set starts — back-left (**Config L1**), front-left (**Config L2**), or front-centre in front of the board (**Config M**). The left gripper picks it up here in all three.
- **Assembly spot**: just right of the middle of the table — the layout board stays here all episode and is never lifted.
- **Layout board**: the working area for the comb, all three ties, the mate, and the wrap. S1 is nearest the connector end, then S2, then S3, then the breakout mark.

- **Parts tray:** the front-right — cable ties, spares, the mating half, and the tape roll, each in its own group. Parts are taken here and returned here.
- **Cutter stand:** the back-right. The cutter stands here whenever it is not in the right gripper.
- **Peg rail:** the right edge at the back. The finished harness hangs here and nothing else does.
- **Left side:** kept clear apart from the wire set, so the left gripper can come in onto the set spot and the board.

Arm assignments

- **Right gripper:** combs the five wires, takes each tie from the parts tray, passes it under the bundle, threads and tensions it, takes and returns the cutter, cuts every tail flush, takes and pushes the mating half, takes and returns the tape roll, wraps the breakout, and takes the right half of the carry to the peg.
- **Left gripper:** picks the wire set off the set spot in all three configs — from the back-left in Config L1, the front-left in Config L2, or the front-centre in Config M — lays it on the board and squares it, holds the bundle closed at each station, holds the connector body during the mate, holds the branch against the main run during the wrap, and takes the left half of the carry to the peg.

Vocabulary

- **Harness:** the finished assembly — one five-wire set, three ties, one mated connector, and one tape wrap.
- **Wire set:** the five loose pre-terminated wires one harness is built from.
- **Set spot:** where the wire set lies at the start of the episode — back-left (**Config L1**), front-left (**Config L2**), or front-centre (**Config M**). One per episode, chosen before recording and never changed mid-episode.
- **Wire:** one insulated conductor in the set.
- **Bundle:** the five wires gathered together and running side by side as one round group.
- **Main run:** the length of the harness where all the wires run together, from the connector to the breakout.
- **Branch:** the two wires that leave the main run at the breakout and carry on to their own free ends.
- **Breakout:** the place where the branch leaves the main run, at the breakout mark on the board. The tape goes here.
- **Free end:** the cut end of a wire, at the end away from the connector.
- **End connector:** the plug already terminated onto the five wires at the left end of the set.
- **Mating half:** the matching part from the parts tray that the end connector pushes into.

- **Latch:** the sprung tab on the end connector that clicks over the mating half when the two are home.
- **Latch up:** the wire set lies on the board with the connector to the left and the latch facing the ceiling. It keeps that position for the whole episode.
- **Square:** the board's front edge lines up with the front edge of the table, so the run reads straight across the board.
- **Station:** one of the three marked places on the main run where a tie goes.
- **S1 / S2 / S3:** the tie station nearest the connector, the one at the middle of the run, and the one just short of the breakout. Tying runs S1, then S2, then S3.
- **Cable tie:** the one-piece strap that closes the bundle at a station.
- **Tie head:** the boxed end of the tie that the tail threads through and locks in.
- **Head up:** the tie head sits on the top of the bundle, facing the camera, before the tie is tensioned.
- **Tie tail:** the loose strap left past the head once the tie is tensioned.
- **Tensioned:** the tie drawn down until no wire slides inside it when the bundle is pushed along the run.
- **Overtight:** the tie pulled down hard enough to press a dent into the insulation or squash the bundle out of round.
- **Loose tie:** a tie the wires still slide inside when the bundle is pushed along the run.
- **Flush cut:** the tail cut off level with the face of the head, leaving no **stub** standing past it.
- **Stub:** a length of tail left standing past the head after a cut that was not flush.
- **Crossover:** a wire that swaps position inside the bundle instead of running straight alongside the others.
- **Stray:** a wire left outside the bundle and not caught by a tie.
- **Slack loop:** a wire standing out of the bundle as a bow because it is longer than the rest.
- **Pulled tight:** a wire drawn straighter than the rest, so it takes the load of the bundle on its own.
- **Round bundle:** the five wires sit together as a round group, not spread out flat as a ribbon.
- **Mated:** the two connector halves are pushed fully together, the latch has clicked over, and their faces meet with no gap.
- **Pull check:** a light, steady pull on the mated connector, straight along the run, for a count of one. The halves must not come apart.
- **Wrap:** one continuous run of tape over the breakout.
- **Half-lap:** each turn of tape covers half the width of the turn before it, so no bare gap is left between turns.
- **Tape tail:** the last unstuck end of the wrap. It is pressed down flat and never left standing.

- **Hung:** the harness hangs on one peg by its S2 tie, connector end down, with the branch hanging free and not looped over the peg.

Steps

Run Steps 1–6 in order on the one layout board, then end the episode with Step 7. Only the first pick in Step 1 depends on the config: the **left gripper** takes the wire set from the back-left in Config L1, the front-left in Config L2, or the front-centre in Config M. Every other line is the same in all three configs.

Step 1: Lay the wire set on the board

Goal: the wire set lies along the layout board, connector end to the left and **latch up**.

Look where the wire set is before reaching for it.

- **IF the wire set is at the back-left (Config L1):** with the **left gripper**, close on the wire set at the **end connector** body, lift it straight up, clear of the set spot, and carry it forward and to the right to the layout board.
- **IF the wire set is at the front-left (Config L2):** with the **left gripper**, close on it at the **end connector** body, lift it straight up, clear of the set spot, and carry it back and to the right to the layout board.
- **IF the wire set is at the front-centre (Config M):** with the **left gripper**, close on it at the **end connector** body, lift it straight up, clear of the set spot, and carry it straight back onto the layout board.

Then, in all three:

- With the **left gripper**, lay it along the run. Never drag or slide the set across the table.
- Lay it with the connector end to the **left** and the **latch up**, with the free ends running out toward the right of the board.
- Open the **left gripper** and lift clear.

Check: all five wires are present, all five run to the one connector, the connector is to the left with the latch up, and no wire is trapped under the board. If the set is the wrong way round, pick it up at the connector with the **left gripper** and lay it again.

Expected state: one wire set on the board, connector left and latch up, and both grippers empty.

Step 2: Comb the five wires into one bundle

Goal: the five wires lie side by side as one **round bundle** along the main run, with no crossover, no stray, no slack loop, and no wire pulled tight.

2.1 Straighten the wires

- With the **left gripper**, hold the wire set down at the **end connector** body so the connector cannot move, and keep it there through the rest of Step 2.
- With the **right gripper**, draw each wire out along the board, one wire at a time, from the connector toward its free end, taking out coil and bend as it goes.
- Do not pull hard enough to move the connector under the **left gripper**.

2.2 Gather the bundle

- With the **right gripper**, bring the wires together against the board so they run side by side from the connector to the breakout mark.
- Work from the connector outward with the **right gripper**, laying each wire alongside the ones already gathered rather than pushing it in among them.
- Keep the three main-run wires together with the **right gripper**, and let the two branch wires fall to the branch side at the breakout mark.

2.3 Take out crossovers and slack

- With the **right gripper**, lift out any wire that swaps position inside the bundle and lay it back so it runs straight alongside the others for the whole main run.
- With the **right gripper**, even out the wires so none stands out as a **slack loop** and none is **pulled tight** against the rest.
- Confirm no wire is left outside the bundle as a **stray**.

Check: all five wires are in the bundle, the bundle is round rather than spread flat, the branch leaves at the breakout mark and nowhere else, and the connector is still to the left with the latch up. If a wire crosses or stands out, comb it back in with the **right gripper** before any tie goes on.

Expected state: one round, combed bundle lying along the run, the left gripper still on the connector, and no tie on the board yet.

Step 3: Tie the bundle at the three stations

Goal: three ties hold the bundle at S1, S2, and S3 — each tensioned, none overtight, every head up, and every tail cut flush.

- Work the stations in this order only: **S1, then S2, then S3**.
- Finish each station completely — tie tensioned and tail cut flush — before starting the next one.
- Turn no tie until the bundle has been combed in Step 2.

3.1 Close the tie at the station

- With the **left gripper**, hold the bundle closed at the station mark so no wire can spring out, and keep it there until the tie is tensioned.
- With the **right gripper**, take one **cable tie** from the parts tray at the front-right.
- With the **right gripper**, pass the tie under the bundle at the station mark and thread the tail through the **tie head**.
- With the **right gripper**, bring the head onto the **top** of the bundle before drawing the tie down.
- With the **right gripper**, draw the tail through until the strap closes on the bundle and no wire slides inside it. Stop as soon as it holds. Do not draw it down into the insulation.
- Open the **left gripper** and lift clear. The bundle must stay closed at that station on its own.

3.2 Cut the tail flush

- With the **right gripper**, take the **flush cutter** from the **cutter stand** at the back-right.
- With the **right gripper**, lay the cutter jaws flat against the face of the **tie head**, with the cut side facing away from the bundle, and cut the tail off level with the head.
- With the **right gripper**, stand the cutter back in the **cutter stand**, handles up, and let go.

3.3 Work S2, then S3

- Repeat 3.1 and 3.2 at **S2**, then at **S3**.
- At **S3**, place the tie short of the **breakout mark**, so it catches only the wires of the main run and never the branch.

Check: after each station — the tie sits on its own station mark, the head is on top of the bundle, no wire slides inside the strap when the bundle is pushed along the run, no dent is pressed into any insulation, the bundle is still round, and no stub stands past the head. If a tie is loose, cut it off with the cutter in the **right gripper**, take a spare from the parts tray, and tie that station again. If a stub stands past the head, take the cutter again with the **right gripper** and cut it level.

Expected state: exactly three ties, one each at S1, S2, and S3, all tensioned, all heads up, all tails cut flush, the cutter standing in its stand, and the connector still to the left with the latch up.

Step 4: Mate the end connector

Goal: the end connector is fully mated to its mating half, the latch has clicked over, and the joint holds a pull check.

- Do not start this step until all three ties are on and checked.

4.1 Bring the halves together

- With the **left gripper**, hold the harness down at the **end connector** body, latch up, and keep it there through the rest of Step 4.
- With the **right gripper**, take the **mating half** from the parts tray at the front-right.
- With the **right gripper**, bring the two halves together in line, keying feature to keying feature, with the latch window over the latch.
- Look at both faces before they meet: no contact bent, pushed back, or missing.

4.2 Push them home

- With the **right gripper**, push the mating half straight onto the connector in one motion until the **latch clicks**.
- Push straight only. The **right gripper** never rocks, twists, or levers the halves together.
- If the halves do not go home in one push, pull them apart with the **right gripper**, line them up square again, and push again. Never force a half that is not lined up.

4.3 Pull check

- With the **right gripper**, pull the mating half straight along the run, away from the harness, light and steady, for a count of one.
- Open the **right gripper** and look. The halves must still be together.
- If they come apart, push them home again with the **right gripper** and pull-check that joint again.
- Open the **left gripper** and lift clear.

Check: the latch stands flat over its window, the two housing faces meet with no gap, the joint held its pull check, and no wire has been pulled out of the back of either housing. If a gap is left, push the halves home again with the **right gripper** and pull-check again.

Expected state: the connector mated and latched, the harness still lying along the run with three ties on it, and both grippers clear.

Step 5: Tape the breakout

Goal: one half-lapped wrap covers the breakout, holds the branch against the main run, covers no tie head, and leaves no tape tail standing.

5.1 Start the wrap

- With the **left gripper**, hold the branch against the main run at the **breakout mark**, and keep it there through the rest of Step 5.
- With the **right gripper**, take the **tape roll** from the parts tray at the front-right.
- With the **right gripper**, start the wrap on the main run on the connector side of the breakout, clear of the **S3** tie head, and press the first turn down onto itself.

5.2 Wrap across the breakout

- With the **right gripper**, wrap toward the branch in a **half-lap**, each turn covering half the width of the turn before it.
- Carry the wrap across the breakout and onto the branch with the **right gripper**, then bring it back over the breakout the same way.
- Keep the tape flat with the **right gripper** — no fold, no wrinkle — and do not stretch it thin over the split.
- Do not let the wrap run over the **S3** tie head or any other tie head.

5.3 Finish the wrap

- Finish the wrap on the main run with the **right gripper**, and lay the **tape tail** down flat against the wrap.
- With the **right gripper**, press the tail down along its whole length so no edge can lift.
- With the **right gripper**, put the tape roll back in the parts tray and let go.
- Open the **left gripper** and lift clear.

Check: the wrap covers the breakout on both sides of the split, the half-lap is even with no bare gap between turns, no tie head is under the tape, no edge of the wrap lifts, and the branch is held against the main run with no wire trapped in a fold. If an edge lifts, press it down again with the **right gripper**. If a tie head is covered, unwind that wrap with the **right gripper** and lay it again clear of the head.

Expected state: one taped breakout, three ties on the bundle, the connector mated, the tape roll back in the parts tray, and both grippers clear.

Step 6: Hang the harness on the peg

Goal: the finished harness hangs on one peg by its S2 tie, connector end down and branch hanging free.

- Do not start this step until the wrap has passed its check.
- With the **left gripper**, close on the harness at the **S2** tie so the bundle takes the weight. With the **right gripper**, close on the harness near the taped breakout.
- Both grippers lift it together, clear of the board, and carry it level to the **peg rail** at the right edge.
- With the **right gripper**, lift the harness over the peg at the **S2** tie and lower it on.
- Let it settle so it hangs by the S2 tie, connector end down, with the branch hanging free and not looped over the peg.
- Open both grippers and lift clear.

Check: the harness hangs on one peg, connector end down, branch hanging free, clear of the table and clear of anything beside it, and it does not slide off the peg when the grippers let go. If

it hangs by a wire instead of the S2 tie, or the branch is looped over the peg, lift it off with both grippers and hang it again.

Expected state: one finished harness hanging on the peg, the layout board bare and still **square** on the assembly spot, and both grippers empty.

Step 7: End the episode

Goal: recording ends with one finished harness hanging on the peg.

1. Confirm the end state:
 - the harness hangs on one peg by its S2 tie, connector end down and branch hanging free;
 - three ties sit at S1, S2, and S3, all tensioned, heads up, tails cut flush, with no stub;
 - the connector is mated, latched, and its two housing faces meet with no gap;
 - the wrap covers the breakout, half-lapped, with no tie head under it and no edge lifting;
 - the cutter stands in its stand, the tape roll is back in the parts tray, and the layout board is bare and square on the assembly spot.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

After each episode

1. Take the harness off the peg. Cut the three ties off, unwind the tape from the breakout, and pull the two connector halves apart.
2. Throw the three cut ties and the used tape away. They are never reused.
3. Put the mating half back in its group in the parts tray, and refill the tray to three ties plus two spares and one tape roll.
4. Comb the wire set straight by hand, check no insulation is split or nicked, and lay it back on the **set spot** for the next episode's config — back-left (Config L1), front-left (Config L2), or front-centre (Config M) — with the connector to the left.
5. Brush tie offcuts and tape scraps off the layout board and off the table, and set the board **square** on the assembly spot.
6. Stand the cutter in the cutter stand, handles up, jaws closed.
7. Run both Setup checklists before the next episode.

At the end of the session

1. Inspect the parts. Replace a wire set with split insulation, a nicked wire, a flattened kink, or a bent contact in its connector. Replace a mating half whose latch window is chipped or whose contacts are bent. Replace the cutter if its jaws no longer close flat or no longer leave a flush cut. Replace the tape roll when it stops sticking to itself.
2. Leave the layout board bare and square on the assembly spot, one wire set on the set spot with the connector to the left, the parts tray filled with three ties plus two spares, one mating half and one tape roll, the cutter standing in its stand, and every peg clear.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config L1 (wire set starts to the left of the board, at the back-left). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the wire set is not in the set spot for the config; a gripper reaches across the table for the wire set, or the right gripper picks it up; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the wire set starts at the back-left, the front-left, or the front-centre and the left gripper picks it in all three; no arm reaches across the table; the IF line followed is the one for the config on the table).

- **Coaching note:** look where the wire set is before the first reach, then follow that config's IF line through Step 1.

Violation: Wire set laid the wrong way

- **Visible cue:** the set ends up on the board with the connector to the right, with the latch facing down or sideways, or lying across the run instead of along it, and the episode moves on.
- **SOP rule broken:** Step 1 (lay the set along the run with the connector end to the left and the latch up, free ends out to the right).
- **Coaching note:** connector left, latch up, every time. Fix it before you comb.

Violation: Wire set dragged or moved off the board

- **Visible cue:** either gripper slides or pushes the set across the table instead of lifting and carrying it, or the layout board is lifted, dragged, or left out of square at the end.
- **SOP rule broken:** Steps 1–6 (lift the set clear and carry it; the layout board stays flat and square on the assembly spot for the whole episode and is never lifted).
- **Coaching note:** bring the wires and the parts to the board. The board never moves.

Violation: Bundle not combed before the first tie

- **Visible cue:** the right gripper takes a tie from the parts tray while wires are still coiled, bent, or spread out across the board, with no comb pass done.
- **SOP rule broken:** Steps 2 and 3 (comb the wires straight and gather them into one round bundle before any tie goes on).
- **Coaching note:** comb first, tie second. A tie freezes whatever mess is under it.

Violation: Bundle left uncombed

- **Visible cue:** at the first tie, or at the end of the episode, a wire swaps position inside the bundle instead of running straight, a wire lies outside the bundle with no tie catching it, a wire stands out as a bow, one wire is drawn straighter than the rest, or the five wires lie spread out flat as a ribbon.
- **SOP rule broken:** Step 2 (lay every wire back so it runs straight, leave no wire outside the bundle, even the lengths so none stands out slack or pulled tight, and keep the bundle round rather than spread flat).
- **Coaching note:** run your eye along the bundle from the connector out. One wire out of line, standing proud, or spread flat means comb again.

Violation: Connector not held while combing

- **Visible cue:** the left gripper is off the connector body while the right gripper draws wires, and the connector slides, spins, or is dragged along the board.

- **SOP rule broken:** Step 2 (the left gripper holds the wire set down at the connector body through the whole of Step 2).
- **Coaching note:** pin the connector first, then pull the wires. Never pull against a loose connector.

Violation: Stations worked out of order or a station skipped

- **Visible cue:** the right gripper ties the stations in some order other than S1, S2, S3, starts a station before the last tie's tail is cut, or the episode ends with fewer or more than three ties.
- **SOP rule broken:** Step 3 (work S1, then S2, then S3, and finish each station — tensioned and cut flush — before starting the next).
- **Coaching note:** one station at a time, connector end first. Tie it, cut it, then move.

Violation: Tie placed off its station

- **Visible cue:** a tie sits away from its printed station mark, or the S3 tie sits over or past the breakout mark and catches a branch wire.
- **SOP rule broken:** Step 3 (put each tie on its own station mark, and place the S3 tie short of the breakout mark so it catches only the main run).
- **Coaching note:** land the strap on the mark. S3 stays on the connector side of the breakout.

Violation: Bundle not held closed while the tie goes on

- **Visible cue:** the left gripper is off the bundle while the right gripper passes or draws a tie, and a wire springs out of the strap or the bundle opens up at that station.
- **SOP rule broken:** Step 3 (the left gripper holds the bundle closed at the station until the tie is tensioned).
- **Coaching note:** close the bundle with the left gripper first, then pass the strap under it.

Violation: Tie tension wrong

- **Visible cue:** the wires slide inside the strap when the bundle is pushed along the run, or the tie moves along the bundle at the check; or the strap has pressed a dent into the insulation, or the bundle is squashed out of round at that station.
- **SOP rule broken:** Step 3 (draw the tail through until the strap closes and no wire slides inside it, and stop as soon as it holds, without drawing it into the insulation).
- **Coaching note:** push the bundle at each station after you tie it. If it slides, cut it off and use a spare. If it dented the wire, you went past the point where it held.

Violation: Tie head not on top

- **Visible cue:** a tie head ends up on the side or the underside of the bundle, or it is tensioned before it is brought onto the top.

- **SOP rule broken:** Step 3 (bring the head onto the top of the bundle before drawing the tie down).
- **Coaching note:** roll the head up on top first. Once it is tensioned, the head stays where it is.

Violation: Tail not cut, or not cut flush

- **Visible cue:** a tie tail is left uncut, is cut leaving a stub standing past the head, or the cutter is laid at an angle and the cut leaves a corner sticking up.
- **SOP rule broken:** Step 3 (lay the cutter jaws flat against the face of the head and cut the tail level with it, leaving no stub).
- **Coaching note:** jaws flat on the head face, cut side out. Look at the head after every cut.

Violation: Cutter not used or not returned

- **Visible cue:** a tail is pulled, twisted, or bitten off with the gripper instead of the cutter, or the cutter is left lying on the board or the table, or it is still in the right gripper during the mate or the wrap.
- **SOP rule broken:** Step 3 (cut tails with the flush cutter only, and stand it back in the cutter stand after each cut).
- **Coaching note:** cutter for tails, stand for the cutter, empty gripper for everything else.

Violation: Wire nicked and the episode continues

- **Visible cue:** the cutter catches a wire while a tail is cut and leaves cut or squashed insulation, and the episode carries on building the harness.
- **SOP rule broken:** Step 3 (cut with the jaws flat against the head face and the cut side facing away from the bundle).
- **Coaching note:** angle the cut away from the wires. A nicked wire ends the build; report it.

Violation: Connector mated before the ties are done

- **Visible cue:** the right gripper takes the mating half from the parts tray while fewer than three ties are on the bundle, or before the ties have been checked.
- **SOP rule broken:** Step 4 (do not start the mate until all three ties are on and checked).
- **Coaching note:** three ties, checked, then the connector. Never the other way round.

Violation: Connector rocked, twisted, or forced

- **Visible cue:** the right gripper wiggles the mating half side to side, turns it, or levers it on, or keeps pushing a half that is not lined up instead of pulling it apart and starting again.
- **SOP rule broken:** Step 4 (push the mating half straight on in one motion; if it does not go home, pull it apart, line it up square, and push again).
- **Coaching note:** straight on, one push. If it fights you, back it off and square it up.

Violation: Connector not fully mated

- **Visible cue:** the latch has not clicked over its window, or a gap is left between the two housing faces, and the episode moves on to the tape.
- **SOP rule broken:** Step 4 (push until the latch clicks, with the two housing faces meeting and no gap left between them).
- **Coaching note:** listen for the click and look at the gap. Both, before you pick up the tape.

Violation: Pull check skipped or not repeated

- **Visible cue:** the right gripper never pulls the mated connector, or the halves come apart during the pull and the episode moves on without them being pushed home and pulled again.
- **SOP rule broken:** Step 4 (pull the mating half straight along the run for a count of one; if it comes apart, push it home again and pull-check that joint again).
- **Coaching note:** every mate earns a pull. Every re-mate earns another one.

Violation: Branch not held during the wrap

- **Visible cue:** the left gripper is off the branch while the right gripper wraps, and the branch lifts away from the main run or a wire is wrapped in a fold of tape.
- **SOP rule broken:** Step 5 (the left gripper holds the branch against the main run at the breakout mark through the whole of Step 5).
- **Coaching note:** hold the branch down first. The wrap only holds what you held.

Violation: Wrap not half-lapped or laid badly

- **Visible cue:** turns of tape leave a bare gap between them, the tape is folded, wrinkled, or stretched thin over the split, or the wrap covers only one side of the breakout.
- **SOP rule broken:** Step 5 (wrap in a half-lap with each turn covering half the one before it, keeping the tape flat, and carry the wrap across the breakout and back).
- **Coaching note:** half on the last turn, half on bare wire, and keep the tape flat as it goes down.

Violation: Tape run over a tie head

- **Visible cue:** the wrap covers the S3 tie head, or any other tie head, at the end of the episode.
- **SOP rule broken:** Step 5 (start the wrap clear of the S3 tie head and never let it run over a tie head).
- **Coaching note:** start on bare bundle between the head and the breakout. Heads stay visible.

Violation: Tape tail left standing

- **Visible cue:** the last end of the tape stands up off the wrap, an edge of the wrap lifts, or the tape roll is left on the board or the table instead of back in the parts tray.
- **SOP rule broken:** Step 5 (lay the tape tail flat against the wrap, press it down along its whole length, and put the tape roll back in the parts tray).
- **Coaching note:** press the tail down its whole length, then park the roll.

Violation: Harness hung wrong or hung early

- **Visible cue:** the harness goes on the peg before the wrap has been checked, hangs by a wire or a free end instead of the S2 tie, hangs connector end up, has the branch looped over the peg, shares a peg, or slides off when the grippers let go.
- **SOP rule broken:** Step 6 (hang only after the wrap check, on one peg, by the S2 tie, connector end down, branch hanging free).
- **Coaching note:** check the wrap, then hang it by S2 with the connector hanging down.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper taking a tie, threading or tensioning a tie, holding the cutter, pushing the mating half, or working the tape roll, or the right gripper being the one that holds the bundle closed at a station.
- **SOP rule broken:** Steps 1–6 (the right gripper combs, ties, cuts, mates, and tapes; the left gripper lays the set and holds the bundle, the connector, and the branch).
- **Coaching note:** right gripper works, left gripper holds. It does not swap.

Violation: Part dropped or knocked off its spot

- **Visible cue:** either gripper drops a tie, the mating half, the tape roll, the cutter, or the harness on the board, the table, or the floor, or tips, pushes, or drags the parts tray, the cutter stand, or the peg rail off its spot.
- **SOP rule broken:** Steps 1–6 (the parts tray, the cutter stand, and the peg rail stay on their spots, and every carry stays level and low over the table).
- **Coaching note:** work slower and lower over the table, and keep each carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the hung harness, the three ties, the mate, and the wrap are confirmed, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 7 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A wire set arrives with split insulation, a cut strand, or a flattened kink, so it cannot be combed straight.
- A wire set arrives with a bent or pushed-back contact in its connector, so it cannot be mated.
- A cable tie will not lock in its own head, or its strap snaps while it is being drawn down.
- The mating half arrives with a broken latch window, so the latch cannot click over.
- The tape roll arrives dried out and will not stick to itself.
- The cutter arrives with jaws that no longer close flat, so no cut can be flush.

Annotation subtasks (from SOP)

1. Lay the wire set on the board, connector left and latch up
2. Hold the connector down with the left gripper
3. Draw each of the five wires straight along the board
4. Gather the wires into one round bundle and take out crossovers and slack
5. Hold the bundle closed at a station with the left gripper
6. Pass a tie under the bundle, thread it, and tension it head up
7. Take the cutter and cut the tie tail flush
8. Repeat the tie and cut at S2, then at S3
9. Take the mating half and push it straight onto the connector until it latches
10. Pull-check the mated connector
11. Hold the branch against the main run and wrap the breakout in a half-lap
12. Press the tape tail down and return the tape roll
13. Lift the harness with both grippers and hang it on the peg by its S2 tie
14. Confirm the end state, return both arms home, and end the episode